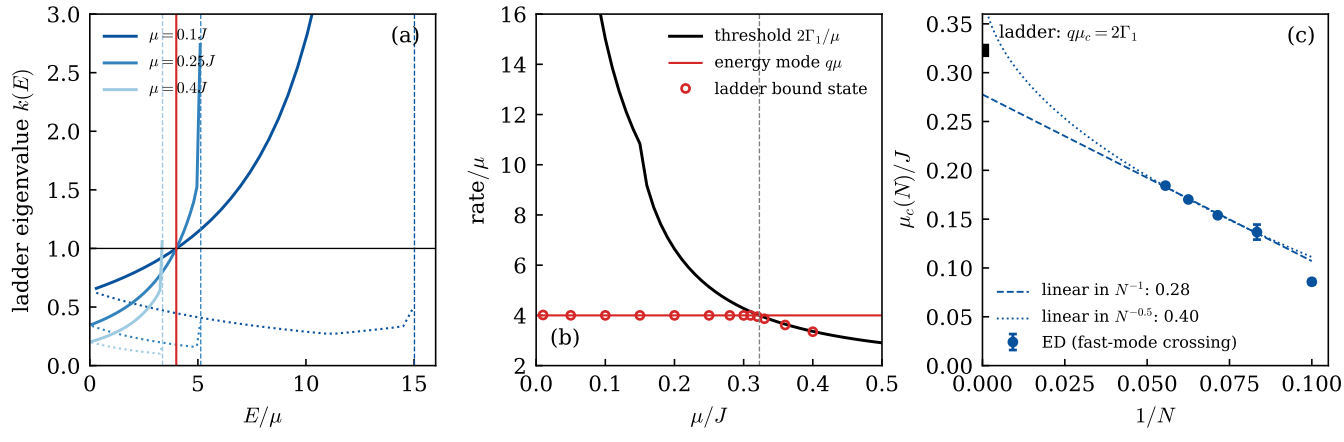
ladder eigenvalue $k(E)$
E/μ
$\mu = 0.1J$
$\mu = 0.25J$
$\mu = 0.4J$
(a)
rate$/\mu$
μ/J
threshold $2\Gamma_1/\mu$
energy mode $q\mu$
ladder bound state
(b)
$\mu_c(N)/J$
$1/N$
ladder: $q\mu_c = 2\Gamma_1$
(c)
linear in N^{-1}: 0.28
linear in $N^{-0.5}$: 0.40
ED (fast-mode crossing)